Project 3 — Discovering Patterns

Describing a dataset with graphs and summaries.

Project 3: Descriptive analysis of a TidyTuesday dataset

Total points: 35 (one-page summary) + 5 (code notebook).

Sample: Project 3 example — Palmer penguins · Sample one-page PDF

In this project you will practice tools from the descriptive statistics section of your statistics toolbox. The purpose of descriptive statistics is summarizing and visualizing data *as they appear in your dataset*, without trying to generalize to a larger population. You can either use a dataset from TidyTuesday or find another one of your choice. The only requirement is that your dataset has at least 4 variables and at least 10 rows.

Once you have chosen four variables, classify each as numerical or categorical. Your code notebook must include appropriate univariate summaries and plots for each of the four variables, and appropriate bivariate summaries and plots for each of the six variable pairs. Your one-page summary includes only the two graphics you judge most interesting, plus at 4 least numeric facts and at least 4 findings drawn from the full exploration.

Project workflow

Phase	What
A — Milestone	Find a dataset; load with `read_csv()` in Colab; run `glimpse()`; choose four variables (name, description, type) — before plots or the one-page summary
B — Full project	Ten plots in the notebook (4 univariate + 6 bivariate), one-page PDF

Phase A instructions and notebook: Project 3 milestone

What to turn in on Canvas (Phase B, after the milestone):

1. One-page summary (file upload): A single PDF (or Google doc Word exported to PDF) that follows the one-page rubric.

2. R code (file upload): A notebook that runs from top to bottom and contains all code for loading the data, all four univariate plots, and all six bivariate plots (10 figures total), with summaries that match each variable type.

If your notebook needs extra files (e.g. a downloaded CSV), zip the notebook and data together or document clearly in the notebook how to obtain the same file.

How to choose and load a TidyTuesday dataset

TidyTuesday publishes one dataset per week on GitHub. Each week lives in its own dated folder under `data/`. If GitHub is new to you, follow the steps below before you write analysis code.

Browse datasets by year

Open the folder for a year, then click a dated subfolder (one week):

Year	Datasets in that year
2018	Weekly folders `2018-04-03`, …
2019	Weekly folders
2020	Weekly folders (includes Palmer penguins, 2020-07-28)
2021	Weekly folders
2022	Weekly folders
2023	Weekly folders
2024	Weekly folders
2025	Weekly folders
2026	Weekly folders

You may also browse curated collections.

Step-by-step: from the list to data in Colab

1. Pick a year from the table and open it on GitHub. You will see many folders named by date (for example `2020-07-28`).
2. Click one folder for a topic that interests you. You land on that week's page (for example 2020-07-28 Palmer penguins).
3. Scroll down and read `readme.md`. Read:
   - the short description at the top (what the data are about);
   - where the data came from (source / citation);
   - the Data Dictionary table (variable names, types, and meanings).
4. In the readme, find the section "Get the data here" (or similar). Under "Or read in the data manually" you will see R code with `read_csv('https://raw.githubusercontent.com/...')`. Copy this line, being sure to include the full URL inside the quotes. This must start with `https://raw.githubusercontent.com/`.
5. In Google Colab (or your `.ipynb`), set the runtime to R: Runtime → Change runtime type → R.
6. In the first code cell, load tidyverse, then paste the `read_csv()` line you copied:

```r
library(tidyverse)

penguins <- read_csv(
  "https://raw.githubusercontent.com/rfordatascience/tidytuesday/main/data/2020/2020-07-28/penguins.csv"

)
```

(Replace the URL with the one from your chosen week.)

7. Inspect the file before plotting:

```r
glimpse(penguins)
head(penguins, 10)
```

8. Use the Data Dictionary to classify columns as numerical or categorical, pick four variables for Project 3, and note any missing values (`sum(is.na(...))` or `summarize(across(...))`).

Note: you may use a non–TidyTuesday source if it has at least 4 columns and 10 rows; still document the URL and load with `read_csv()` when the file is a CSV.

Steps for Phase A (milestone)

Complete Phase A in the milestone notebook in Colab before any plots or the one-page summary. Project 3 milestone.

Step	Task
A1	Choose a TidyTuesday week or other dataset. Copy the location (source) and one-sentence description.
A2	Load the CSV in Colab (R): `library(tidyverse)` + `read_csv("https://raw.githubusercontent.com/...")` from the readme.
A3	Run `glimpse()` and `head()` to confirm the file loaded correctly
A4	From all columns, choose four variables for Project 3. For each: name, description (from readme), type (categorical or numerical).

Optional: submit the milestone notebook to the Project 3 — Milestone assignment in Canvas.

Steps for Phase B (full project)

Use the same dataset and four variables from Phase A.

1. In your notebook, summarize each of your four variables appropriately:
   - Categorical: counts and proportions and a bar chart or pie chart (one plot per variable).

   - Numerical: at least one measure of center and one of spread (e.g. mean and SD, or median and IQR), plus one plot such as a histogram, dot plot, or boxplot (one plot per variable).

   This gives four univariate plots in the notebook.
2. In your notebook, explore every pair among your four variables (six pairs total). Use the bivariate method from class that matches the variable types:

Pair types	Graph / table
Numerical × numerical	Scatterplot (and Pearson r if roughly linear)
Numerical × categorical	Side-by-side boxplots (or grouped histograms)
Categorical × categorical	Stacked bar chart (counts or 100% fill) and two-way table

   Each pair of the four variables gets one primary bivariate plot in the notebook for six bivariate plots total.
3. Pick the two most interesting plots from your notebook for your one-page summary. Each must have a clear title and axis or legend labels (with units if applicable). The other eight plots stay in the notebook only.
4. Write the one-page summary (rubric below). It must include your two chosen plots, at least four clearly labeled numeric summaries (e.g. two means and two SDs, or medians and IQRs, or counts/proportions expressed as numbers: enough that a reader sees four distinct quantitative facts), and a short paragraph of your four most interesting findings.
5. Check that you can answer five Canvas short questions, review and update your report as needed and then upload the one-page PDF and full notebook to Canvas.

Project 3 — Check yourself

1. Which dataset did you use? Briefly give the context of this dataset.

2. How many variables (columns) are in the full dataset?

3. Which four variables did you choose? What are their types?

4. How many observations (rows) are in the full dataset?

5. Was there missing data in any of your four chosen variables? If yes, say which variable(s) and how you handled or reported it.

One-page summary rubric (35 points total)

Points	Criterion
8	Short paragraph: what the dataset is, where it came from, a research or exploratory question, how many variables and how many observations (in the full dataset).
8	Two graphs, chosen as the most informative/interesting out of the ten you created in the notebook; each has an informative title and axis/legend labels (units where needed).
8	At least four numerical summaries with clear labels (and units if applicable)—in a small table or woven into sentences. Summaries should match variable types (e.g. proportions for categorical variables; mean/median and spread for numerical variables).
8	Short paragraph describing four interesting findings from your four variables (and/or their relationships).
3	Format: One side of a standard page (8.5×11 in.), complete sentences.

Code notebook rubric (5 points)

A code notebook (.ipynb) that runs from top to bottom and includes:

- `library(tidyverse)` and `read_csv("https://raw.githubusercontent.com/...")` for your dataset (same URL as in your TidyTuesday readme, or documented alternative).

- `glimpse()` (or similar) showing you loaded the correct file.

- Appropriate summaries and one plot for each of your four variables (4 univariate plots).

- Appropriate summaries and one plot for each of the six pairs among those four variables (6 bivariate plots).

- Only two of these ten plots appear on your one-page summary; the notebook shows your complete exploration.